

Problem

Design an active low-pass filter with dc gain of 0.25 and a corner frequency of 500 Hz.

Step-by-step solution

Step 1 of 1

$$DC \text{ gain} = \frac{R_f}{R_i} = \frac{1}{4} \rightarrow R_i = 4R_f$$

$$\text{corner frequency} = \omega_c = \frac{1}{R_f C_f} = 2\pi(500) \text{ rad/s}$$

If we select $R_f = 20K\Omega$

then $R_i = 80K\Omega$

and $C = 15.915nF$